Virtual Identification

Virtual Identification + CampX together are basically for Indian education. Below is the detailed, developer-ready table you asked for: features, pages, and which role can do what (Student/Parent App; Teacher Admin; Institutional Admin; Super Admin/Developer).

Legend for roles (bold letters only)

- **S/P** = Student / Parent
- **T** = Teacher / Faculty
- **H** = Head officials (HOD, Principal, In-charges)
- **A** = School / College Admin
- **D** = Super Admin / Developer

"Access" column meaning:

- View = can only see
- Create = can add
- Edit = can update
- Delete = can remove
- Manage = full control (create+edit+delete+approve)

Main Features vs Detailed Pages vs Roles

S. No.	Feature / Module	Detailed Pages (what you will build)	Roles (who can use it and how)
1	Login & Security	Login, Forgot Password, Reset Password, Change Password, Logout, Active Sessions / Devices	**S/P**: login, forgot/reset, change own password, view/revoke own devices (view/edit own). **T**: same for self. **H**: same for self. **A**: same for self + reset passwords for students/teachers. **D**: same for self + create first admin account for each institution.

2	Profiles & Virtual ID	Student Profile, Parent Profile, Teacher Profile, Admin Profile, Developer Profile, Virtual ID (QR/Face), Profile Edit	**S/P**: view/edit own profile, view own Virtual ID. **T**: view/edit own profile, view students' basic profile + QR (view-only). **H**: view all profiles in their department. **A**: create/edit all user profiles, generate Virtual IDs. **D**: set institution details and branding.
3	Role & Permission Control	Role List, Assign Role, Permission Matrix, Role Change History	**S/P**: view own role only. **T**: view own role only. **H**: view department roles list. **A**: assign roles (teacher, HOD, class teacher, fee manager) for their institution. **D**: design default role templates for all institutions.
4	Privacy & Consent	Privacy Settings, Biometric Consent, Data Download Request	**S/P**: accept/revoke biometric consent, set privacy options. **T**: same for self. **H**: view consent status of students in department. **A**: manage institutional rules about retention and consent. **D**: global privacy defaults.
5	Smart AI Attendance	Student Attendance Dashboard, Class Attendance Dashboard, Attendance Detail per Student, Face Enrollment, Flagged Attendance, Attendance	**S/P**: view own attendance calendar and charts (view only). **T**: view class attendance, trigger manual mark for their own class, send override

		Override Requests	requests. **H**: approve/reject overrides, view department attendance. **A**: configure attendance rules, manage sessions and attendance data. **D**: configure AI engine and thresholds.
6	CCTV / Kiosk Attendance Admin	CCTV Live View (admin), Kiosk List, Kiosk Settings (location, rules)	**S/P**: no access. **T**: no access. **H**: read-only view of feeds for their department (optional). **A**: manage kiosks, link to classes, see health status. **D**: setup integration per institution.
7	Attendance Rules & Policies	Attendance Policy (late rules, min time), Holiday / No-Class Setup	**S/P**: view rules only. **T**: view rules. **H**: suggest policy changes (request). **A**: define rules for institution. **D**: define global limits (max/min boundaries).
8	Attendance Notifications	Attendance Alert Settings, Attendance Notification Log	**S/P**: view notifications received. **T**: view alerts sent for their classes. **H**: view department alerts. **A**: configure when and how alerts go to parents/students. **D**: manage default templates

			and languages.
9	AI Smart Timetable	My Timetable (student/teacher), Class Timetable, Timetable Generator, Timetable Conflict Resolver, Timetable Change History	**S/P**: view own timetable only. **T**: view own timetable, request swaps. **H**: approve timetable changes for department, view department timetable. **A**: generate full timetable, solve complex conflicts. **D**: tune algorithm settings.
10	Event & Academic Calendar	Institution Calendar, My Calendar, Event Details, Event Management	**S/P**: view My Calendar, view events and reminders. **T**: view calendar, create class events (within scope). **H**: create department events, approve teacher events. **A**: manage institution-wide events and holidays. **D**: none (optional overview only).
11	Fee Plans & Invoices	Fee Plan Setup, Invoice List, Fee Waiver / Scholarship, Due List	**S/P**: no access to setup; only see own dues (see below). **T**: cannot view fee amounts; optional “fee clear / not clear” flag only. **H**: view collection summary for department (no student-level amounts if you want). **A**: create fee plans, generate invoices, manage waivers and concessions. **D**: select payment gateway per institution.

12	Fee Payment Portal	Fee Payment Home, Payment Methods, Payment Confirmation, Receipt Download, Transaction History	**S/P**: view outstanding fees, pay online, download receipts, see history. **T**: no direct access. **H**: no direct access. **A**: read-only view of payments here (admin has finance dashboards separately). **D**: none.
13	Transactions & Reconciliation	Online Transactions List, Manual Payment Entry, Failed Transactions & Retry, Refund Management	**S/P**: view own transactions only. **T**: no access. **H**: view aggregated numbers for their department only. **A**: reconcile payments, record cash/cheque, manage refunds. **D**: view system-wide error patterns.
14	LMS – Courses & Content	My Courses, Course Detail, Lessons & Topics, Resources / Handouts	**S/P**: view courses enrolled, view resources and lesson plans. **T**: create and manage course content, upload resources. **H**: monitor content quality in department. **A**: enable/disable LMS per course/grade. **D**: manage storage and size limits.
15	Assignments & Submissions	Assignment List, Assignment Detail, Submit Assignment,	**S/P**: view assignments, submit files, view grades and comments.

		Grade & Feedback	**T**: create assignments, view submissions, grade and give feedback. **H**: view assignment stats per course or department. **A**: view institution-level load and status. **D**: none.
16	Homework & AI Helper	Homework List, AI Homework Helper, AI Explanation View, Homework Progress	**S/P**: see homework, chat with AI for hints and explanations (no direct copy answers), see completion progress. **T**: create homework, see completion analytics and AI-usage signals. **H**: track homework load vs performance. **A**: set AI helper rules (what is allowed). **D**: integrate models and guardrails.
17	Academic Chatbot	Ask Assistant, Chat History, Escalate to Human Support	**S/P**: ask for homework, attendance, exam doubts, guidance. **T**: ask about schedule, class analytics, basic queries. **H**: ask about department performance, risk alerts. **A**: ask admin/system queries; view usage charts. **D**: configure knowledge base and intents.
18	Exam Scheduling & Timetable	Exam Calendar, Exam Timetable, Exam Setup, Hall Ticket (optional)	**S/P**: view exam timetable (and hall ticket if you add).

			T: view their invigilation/assessment schedule. **H**: configure exams for their department. **A**: configure full exam structure for institution. **D**: none.
19	Marks Entry & Results	Marks Entry, Result Verification (HOD), Publish Results, Student Result Card	**S/P**: view own result cards and history. **T**: enter marks for their subjects, view class performance. **H**: verify and approve results for department. **A**: final publish and control revaluation / retotaling process. **D**: none.
20	AI Test & Practice Engine	Practice Tests List, Start AI Test, Test Result & Analysis, Teacher Test Request	**S/P**: take practice tests, see strengths/weaknesses. **T**: request tests per syllabus and difficulty, assign to classes. **H**: observe test usage and outcomes. **A**: enable/disable for classes or grades. **D**: manage question generation models and filters.
21	Performance Analytics	Student Analytics, Class Analytics, Department Analytics, Institution Analytics, At-Risk Students	**S/P**: see own performance trends and insights. **T**: see analytics per class, list of at-risk students with

			suggestions. **H**: see department-level analytics and early warning lists. **A**: institution-wide dashboards (attendance, exams, fees, engagement). **D**: configure analytics models and metrics.
22	Digital Noticeboard	Feed (timeline), Announcements (official), Notice Details, Create / Schedule Notice	**S/P**: view feed + announcements; filter by type (general, academics, opportunities, events). **T**: create class / subject notices; edit their own notices. **H**: approve or broadcast department notices, pin important posts. **A**: manage institution-level notices and categories. **D**: design global notice categories.
23	Messaging & PTM	Messages (Inbox/Outbox), Class Channel, Parent–Teacher Chat, PTM Scheduler, PTM Slot Booking	**S/P**: send/receive messages with teachers, book PTM slots. **T**: send messages to class/parents, schedule PTM slots, record PTM outcomes. **H**: view department PTM schedule and summaries. **A**: define communication rules and monitor abuse reports. **D**: none.

24	Clubs & Co-Curricular	Clubs Directory, My Clubs, Club Detail, Co-Curricular Calendar, Club Admin	**S/P**: browse, join clubs, see events. **T**: manage clubs where they are mentor. **H**: approve club activities for department. **A**: approve new clubs and see engagement analytics. **D**: none.
25	Tools Hub	Tools Home (quick links to Attendance, Fees, LMS, Events, Clubs, Analytics, Chatbot)	**S/P**: fast access to student tools. **T**: fast access to teacher tools. **H**: fast access to departmental tools. **A**: fast access to admin tools. **D**: fast access to platform tools.
26	Feedback System	Submit Feedback, Feedback List (for admins/teachers), Feedback Analytics	**S/P**: submit feedback on courses/teachers/system (optionally anonymous), check status of non-anonymous feedback. **T**: see feedback about their courses (aggregated), respond where allowed. **H**: view feedback analytics for department. **A**: view institution feedback trends and actions. **D**: design feedback forms and scales.
27	Helpdesk & Support	Create Ticket, Ticket Status, FAQ / Knowledge Base, Contact Support	**S/P**: raise tickets (login, attendance, fee, academic). **T**: raise staff tickets.

			H: raise department tickets. **A**: manage and assign tickets inside institution. **D**: manage platform-level tickets.
28	Notification Center	Notification List, Notification Settings (channel, frequency), Template Management (admin)	**S/P**: see notifications, set preferences (where allowed). **T**: same. **H**: see department notifications summary. **A**: create and manage notification templates, set rules. **D**: configure providers and languages.
29	Audit & Activity Logs	User Activity Log (per user), System Audit Log (admin)	**S/P**: see own logins and important actions. **T**: see own actions (marks entries, notices). **H**: view logs for department actions. **A**: view all institution logs and filter by feature. **D**: view system-wide audits.
30	Institution Management	Institution List, Institution Detail & Settings, License & Limits, Data Backup / Export	**S/P**: no access. **T**: no access. **H**: no access. **A**: view own institution settings summary. **D**: create institutions, set module access, generate admin credentials, run backups.

Complete List of Detailed Pages (for routing / screen list)

This is just a flat list so you can directly map to routes/screens.

S. No.	Detailed Page Name
1	Login
2	Forgot Password
3	Reset Password
4	Change Password
5	Logout (confirm screen)
6	Active Sessions / Devices
7	Student Profile
8	Parent Profile
9	Teacher Profile
10	Admin Profile
11	Developer Profile
12	Virtual ID (QR / Face)
13	Profile Edit

14	Role List
15	Assign Role
16	Permission Matrix
17	Role Change History
18	Privacy Settings
19	Biometric Consent
20	Data Download Request
21	Student Attendance Dashboard
22	Class Attendance Dashboard
23	Attendance Detail per Student
24	Face Enrollment
25	Flagged Attendance List
26	Attendance Override Requests
27	CCTV Live View (admin)
28	Kiosk List
29	Kiosk Settings
30	Attendance Policy

31	Holiday / No-Class Setup
32	Attendance Alert Settings
33	Attendance Notification Log
34	My Timetable (Student)
35	My Timetable (Teacher)
36	Class Timetable View
37	Timetable Generator
38	Timetable Conflict Resolver
39	Timetable Change History
40	Institution Calendar
41	My Calendar
42	Event Details
43	Event Management
44	Fee Plan Setup
45	Invoice List
46	Fee Waiver / Scholarship
47	Fee Due List

48	Fee Payment Home
49	Payment Methods
50	Payment Confirmation
51	Receipt Download
52	Transaction History (Student/Parent)
53	Online Transactions List (Admin)
54	Manual Payment Entry
55	Failed Transactions & Retry
56	Refund Management
57	My Courses
58	Course Detail
59	Lessons & Topics
60	Resources / Handouts
61	Assignment List (Student)
62	Assignment List (Teacher)
63	Assignment Detail
64	Submit Assignment

65	Grade & Feedback (Teacher)
66	Homework List
67	AI Homework Helper
68	AI Explanation View
69	Homework Progress
70	Ask Assistant (Chatbot)
71	Chat History
72	Escalate to Human Support
73	Exam Calendar
74	Exam Timetable
75	Exam Setup
76	Hall Ticket (optional)
77	Marks Entry
78	Result Verification (HOD)
79	Publish Results
80	Student Result Card
81	Practice Tests List

82	Start AI Test
83	Test Result & Analysis
84	Teacher Test Request
85	Student Analytics (self view)
86	Class Analytics
87	Department Analytics
88	Institution Analytics
89	At-Risk Students List
90	Feed (Noticeboard Timeline)
91	Announcements List
92	Notice Details
93	Create / Schedule Notice
94	Messages Inbox
95	Messages Outbox
96	Class Channel
97	Parent–Teacher Chat
98	PTM Scheduler

99	PTM Slot Booking
100	PTM Summary / Notes
101	Clubs Directory
102	My Clubs
103	Club Detail
104	Co-Curricular Calendar
105	Club Admin Page
106	Tools Home
107	Submit Feedback
108	Feedback List (Teacher/Admin)
109	Feedback Analytics
110	Create Ticket (Helpdesk)
111	Ticket Status
112	FAQ / Knowledge Base
113	Contact Support
114	Notification List
115	Notification Settings

116	Notification Template Management
117	User Activity Log (per user)
118	System Audit Log (admin)
119	Institution List (Developer)
120	Institution Detail & Settings
121	License & Limits
122	Data Backup / Export